

Pratt & Whitney R-1830 Twin Wasp

The Twin Wasp (known as the R-1830 on military aircraft) is the most manufactured aircraft engine of all time, with more than 178,000 built. Buick-made versions powered most of the Consolidated B-24 Liberator bombers used during World War II. The design of this iconic engine was initiated in 1931, yet it remained in production into the 1950s.

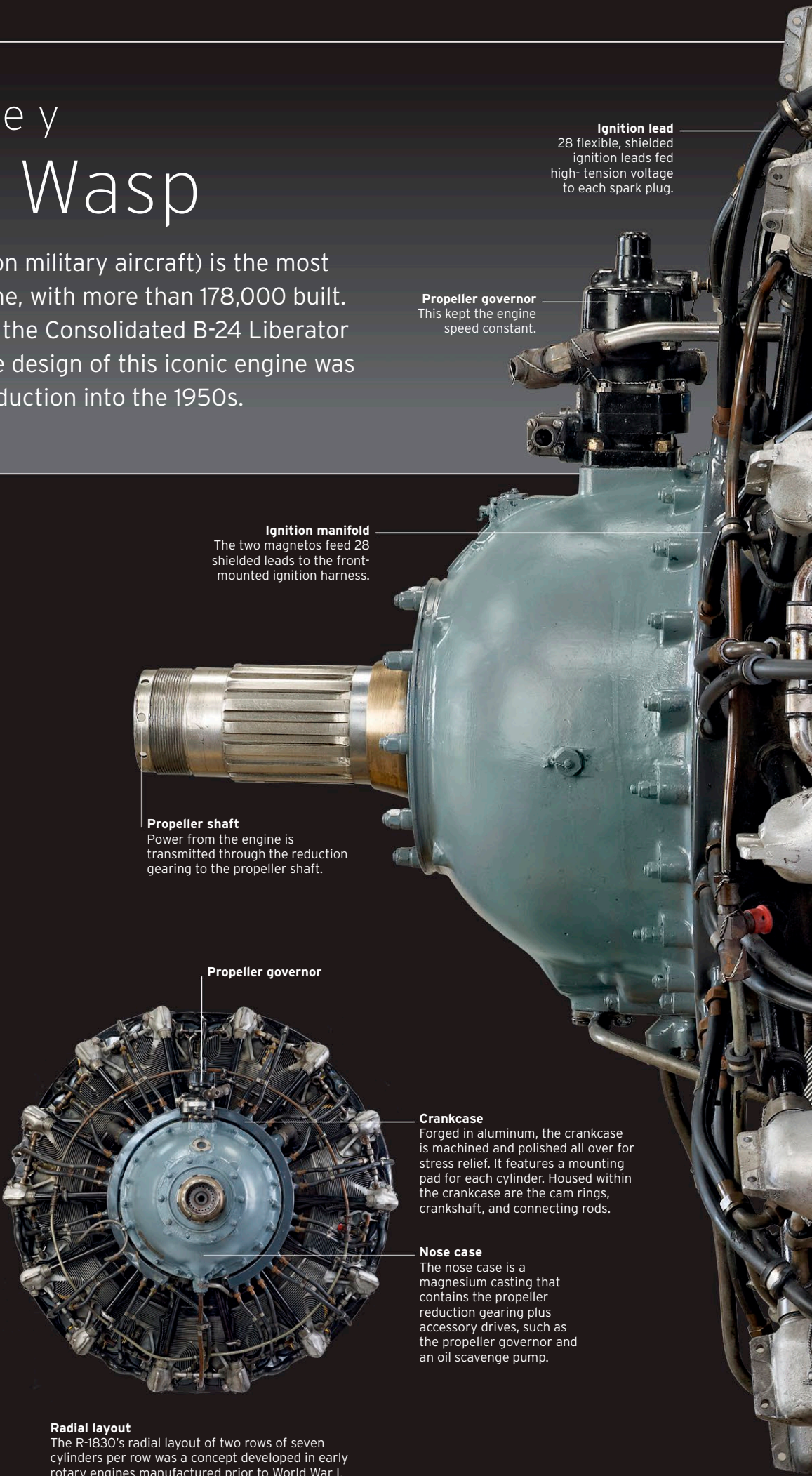
SUPERCHARGED ENGINE

This engine was configured as an air-cooled, 14-cylinder, two-row radial. All Twin Wasps were supercharged—some had two-speed supercharging and others augmented the gear-driven supercharger with an exhaust-driven turbo-supercharger. The massive crankshaft was supported in equally massive roller bearings and the connecting rod bearings ran in sterling silver—a common bearing material for World War II aircraft engines developed by Pratt & Whitney.

ENGINE SPECIFICATIONS	
Dates produced	1932-late 1950s
Configuration	Air-cooled 14-cylinder 2-row radial engine
Fuel	115/145 Grade avgas
Power output	800-1,450 hp @ 2,700 rpm
Weight	1,438 lb (652 kg) (typical)
Displacement	1,830 cu in (29.988 liters)
Bore and stroke	5.5 in x 5.5 in (14 cm x 14 cm)
Compression ratio	6.7:1



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Ignition lead
28 flexible, shielded ignition leads fed high-tension voltage to each spark plug.

Propeller governor
This kept the engine speed constant.

Ignition manifold
The two magnetos feed 28 shielded leads to the front-mounted ignition harness.

Propeller shaft
Power from the engine is transmitted through the reduction gearing to the propeller shaft.

Propeller governor

Crankcase
Forged in aluminum, the crankcase is machined and polished all over for stress relief. It features a mounting pad for each cylinder. Housed within the crankcase are the cam rings, crankshaft, and connecting rods.

Nose case
The nose case is a magnesium casting that contains the propeller reduction gearing plus accessory drives, such as the propeller governor and an oil scavenge pump.

Radial layout
The R-1830's radial layout of two rows of seven cylinders per row was a concept developed in early rotary engines manufactured prior to World War I.